

Q1 Boolean laws

1 Point

$$\rightarrow X \cdot (\underbrace{Y + \bar{Y}}_{1}) \rightarrow X \cdot 1$$

For proving boolean equation $X \cdot Y + X \cdot \bar{Y} = X$, which of the following boolean laws must be used?

$$A + 0 = 0$$

$$A + 1 = A$$

$$A \cdot (B + C) = A \cdot B + A \cdot C$$

distribution

$$A + \bar{A} = 1$$

complement

$$A \cdot \bar{A} = 0$$

Q2 Prime detector

5 Points

We implement a prime number detector as a combinatorial circuit. This circuit takes a 3-bit input (b_2, b_1, b_0) which represents an unsigned base 2 integer with b_0 being the LSB. It outputs a single bit O indicating whether the input is a prime number or not.

Q2.1 Truth table

1 Point

How many total rows does the truth table corresponding to the 3-bit prime number detector circuit have? To facilitate autograding, please write a single number.

8

$$2^3 = 8$$

Q2.2 Truth table

1 Point

Select all rows of the truth table that correspond to the output bit $o = 1$. (Note: Neither 0 nor 1 is a prime number). Note the input columns of the truth table are ordered as $b_2 b_1 b_0$.

(Note: Neither 0 nor 1 is a prime number)

0 1 2 3 4 5 6 7
✓ ✓ ✓ ✓

000

001

010

011

100

101

110

111

Q2.3 Sum of products

1 Point

Given the truth table, a combinatorial circuit can be built as a sum of products where each product term corresponds to a row in Q2.2. Suppose the row with input $b_2 b_1 b_0 = (101)_2$ has output $o = 1$, select the product term that corresponds this row.

$$\begin{array}{c} \downarrow \downarrow \downarrow \\ b_2 \cdot \overline{b_1} \cdot b_0 \rightarrow \underline{1} \end{array}$$

$$b_2 \cdot \overline{b_1} \cdot \overline{b_0}$$

$$\overline{b_2} \cdot b_1 \cdot \overline{b_0}$$

✓ $b_2 \cdot \overline{b_1} \cdot b_0$

$$b_2 \cdot b_0$$

$$b_1$$

None of the above

Q2.4 ROM

1 Point

If you are to using a ROM to implement the prime number detector circuit. What is the minimal size of the ROM required? To facilitate autograding, please write your answer as the product of two numbers, $h \cdot w$, where h is the ROM's height and w is the ROM's width.

8*1

h : number of entries
 w : number of output

Q2.5 ROM

1 Point

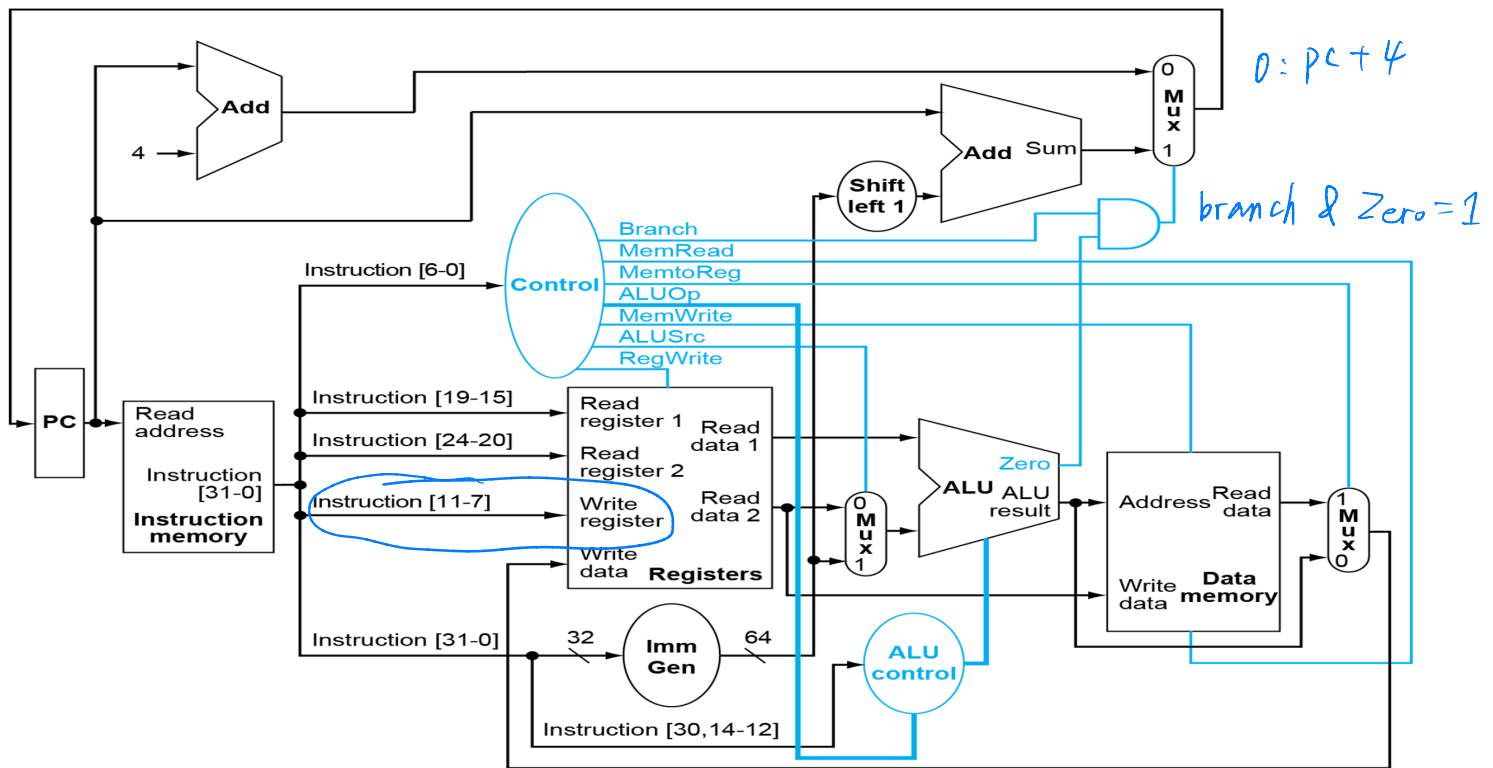
What is the value of the ROM entry at index or address $(101)_2$?

1

Q3 Basic Non-pipelined CPU

4 Points

This question is based on the single-cycle non-pipelined CPU design, shown as follows:



Q3.1 Data path

1 Point

$$x_6 = x_7 + x_8$$

Suppose the RISC-V instruction being executed is `add x6, x7, x8`, where x6 is the destination register, and x7, x8 are the 1st, 2nd source register operand, respectively.

What are the values corresponding to `Instruction[11-7]` that are fed into the "write register" pins of the RegisterFile?

To facilitate autograding, please write your answer in exactly 5 binary digits, with higher order bits on the left.

$$(6)_{10} = (00110)_2$$

00110

Q3.2 Control path

1 Point

There are 3 Mux in the Figure whose selectors are to be set by the control logic, located at the top-right, bottom-middle, and bottom-right.

a branching instruction

Which of the following instruction(s) correspond to the top-right Mux's selector being set to 1?

Note: In the figure, each mux has labelled input signal whose label is either '0' or '1'. The input's label indicates which value the selector should be set in order to select that specific input. Be careful that '0' and '1' labels for the 3 muxes are not always in the same location.

`add x6, x7, x8` //x6 = x7+x8

`beq x6, x7, 100` //jump to PC+2*100 if x6 and x7 are equal.

`ld x5, 40(x6)` //load a doubleword (8-byte) from *Memory*[x6 + 40] to register `x5`

`addi x6, x7, 200` //x6 = x7+200

`sd x5, 40(x6)` //store a doubleword (8-byte) from register `x5` to *Memory*[x6 + 40]

Q3.3 Control path

1 Point

Continuing from Q3.2, which of the following instruction(s) correspond to the bottom-middle Mux's selector (aka `ALUSrc`) being set to 1?

`add x6, x7, x8`

`beq x6, x7, 100`

`ld x5, 40(x6)`

`addi x6, x7, 200`

`sd x5, 40(x6)`

0: register

1: immediate

$x6 + 40$

$x7 + 200$

$x6 + 40$

Q3.4 Control path

1 Point

Continuing from Q3.2 and Q3.3, which of the following instructions correspond to the bottom-right Mux's selector (aka `MemToReg`) being set to 1?

`add x6, x7, x8`

`beq x6, x7, 100`

`ld x5, 40(x6)`

`addi x6, x7, 200`

`sd x5, 40(x6)`

1: read memory

0: write memory